ECSE 597 Circuit Simulation and Modelling

Assignment 1

Due on Feb 24th, 2026

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1 Netlist Reader

In this section, 8 functions are created in MatLab to add the appropriate stamp of the particular component to the MNA equations.

1.1 Current Source

Current source stamp is created such that
if there is a current source between ground and node i

$$\mathbf{b} = \begin{bmatrix} \vdots \\ I_a \\ \vdots \end{bmatrix}$$

If there is a current source between node i and node j

$$\mathbf{b} = \begin{bmatrix} \vdots \\ I_a \\ \vdots \\ -I_a \\ \vdots \end{bmatrix}$$

The code for this function is displayed below:

```
function cur(n1,n2,val)
global b
if n1 ~= 0
    b(n1,1) = b(n1,1) + val;
end
if n2 ~= 0
    b(n2,1) = b(n2,1) - val;
 end
end % end function
```

1.2 Voltage Source

For the voltage source stamp both G and b are being modified. An extra new row is being added. An extra new row and column is added to G matrix and an extra new row is added to the RHS b matrix. For a voltage source between node i and j. The stamp matrix is shown below. If its connected to ground, the corresponding term is simply zero.

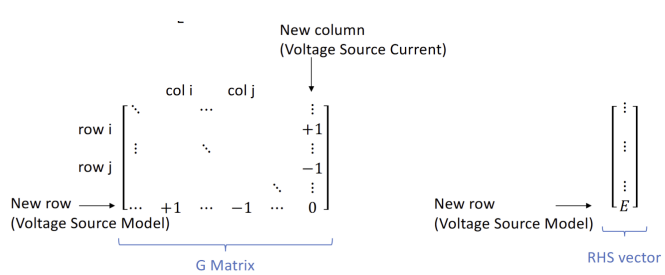

Figure 1: Voltage Source Stamp Formula

The code is displayed below:

```
function vol(n1,n2,val)

global G
global b
global C

d = size(G,1);      % current size
xr = d+1;           % new row index

b(xr) = 0;          % expand b
G(xr,xr) = 0;       % expand G
C(xr,xr) = 0;       % expand C

if (n1 ~= 0)
    G(n1,xr) = G(n1,xr) + 1;
    G(xr,n1) = G(xr,n1) + 1;
end

if (n2 ~= 0)
    G(n2,xr) = G(n2,xr) - 1;
    G(xr,n2) = G(xr,n2) - 1;
end

b(xr) = b(xr) + val;

end %end of function
```

1.3 Resistor

For a resistor stamp between node i and node j or between noe i and ground, the G matrix is modified using the formula below:

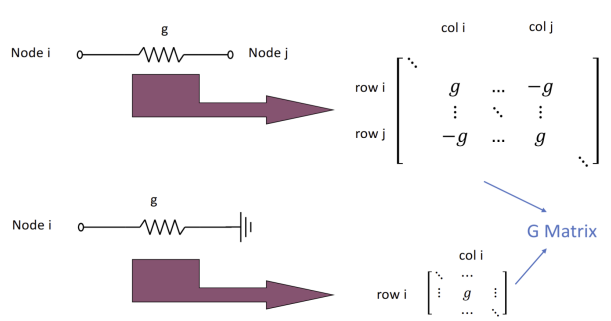

Figure 2: Resistor Stamp Formula

The code is displayed below:

```
function res(n1,n2,val)
global G
g = 1/val;
```

```
if n1 ~= 0
    G(n1,n1) = G(n1,n1) + g;
end

if n2 ~= 0
    G(n2,n2) = G(n2,n2) + g;
end

% Add off-diagonal terms
if n1 ~= 0 && n2 ~= 0
    G(n1,n2) = G(n1,n2) - g;
    G(n2,n1) = G(n2,n1) - g;
end %end of if

end %end function
```

1.4 Inductor

For an inductor, both G and C matrix are being modified. An extra new row and column are created respectively. If the node is connected to ground, the corresponding position in G matrix is zero and C matrix stays the same.

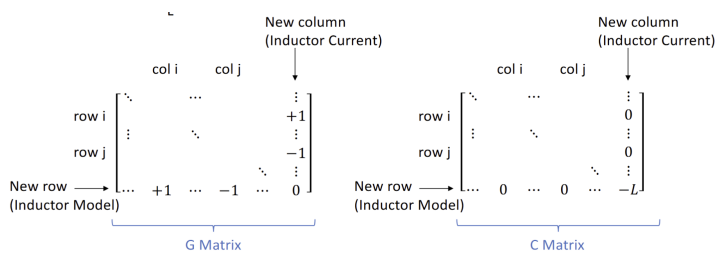

Figure 3: Inductor Stamp Formula

The code is shown below:

```
function ind(n1, n2, val)
global G
global b
global C

d = size(G,1);      % current size
xr = d+1;           % new row index

b(xr) = 0;          % expand b
G(xr,xr) = 0;       % expand G
C(xr,xr) = 0;       % expand C

if (n1 ~= 0)
    G(n1,xr) = G(n1,xr) + 1;
    G(xr,n1) = G(xr,n1) + 1;
```

```
end

if (n2 ~= 0)
    G(n2,xr) = G(n2,xr) - 1;
    G(xr,n2) = G(xr,n2) - 1;
end

C(xr, xr) = C(xr, xr) - val;

end % end of function
```

1.5 capacitor

For capacitor, only matrix C is modified.

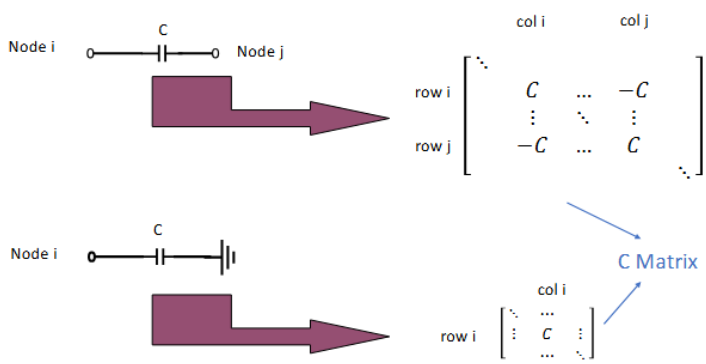

Figure 4: Capacitor Stamp Formula

The code is shown below:

```
function cap(n1,n2,val)
global C
% Add to diagonal
if n1 ~= 0
    C(n1,n1) = C(n1,n1) + val;
end

if n2 ~= 0
    C(n2,n2) = C(n2,n2) + val;
end

% Add off-diagonal terms
if n1 ~= 0 && n2 ~= 0
    C(n1,n2) = C(n1,n2) - val;
    C(n2,n1) = C(n2,n1) - val;
end

end
```

1.6 Voltage Controlled Current Source

For VCCS, G matrix is being modified. It follows the formula below:

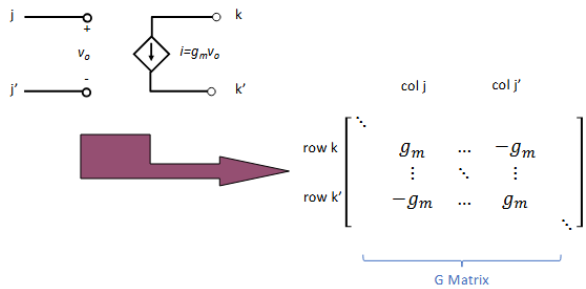

Figure 5: VCCS Formula

```
function vccs(nd1, nd2, ni1, ni2, val)
%dependent nodes nd1, nd2
%independent nodes ni1, ni2
%I nd1 to nd2 = val*(Vni1-Vni2)
global G
if nd1 ~= 0
    if(ni1 ~= 0)
        G(nd1, ni1) = G(nd1, ni1) + val;
    end
    if(ni2 ~= 0)
        G(nd1, ni2) = G(nd1, ni2) - val;
    end
end %end nd1 != 0
if nd2 ~= 0
    if(ni1 ~= 0)
        G(nd2, ni1) = G(nd2, ni1) - val;
    end

    if(ni2 ~= 0)
        G(nd2, ni2) = G(nd2, ni2) + val;
    end
end % end nd2 != 0
end %end function
```

1.7 Voltage Controlled Voltage Source

The VCVS stamping requires the modification of G matrix.

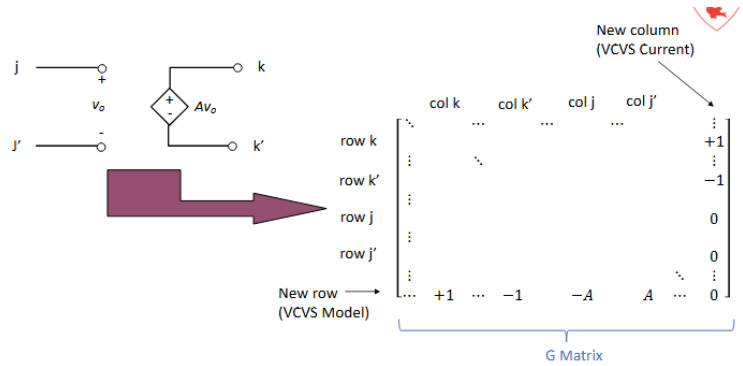

Figure 6: VCVS Formula

The code is shown below:

```
function vcvs(nd1, nd2, ni1, ni2, val)
%note for vccs use as op am
% nd1 would be k which is 0
% and nd2 would be out which is kprim
% ni1 would be + end ni2 would be -end
global G
global b
global C
d = size(G,1);
xr = d+1;
% Expand system
b(xr) = 0;
G(xr,xr) = 0;
C(xr,xr) = 0;

if nd1 ~= 0
    G(nd1,xr) = G(nd1,xr) + 1;
    G(xr,nd1) = G(xr,nd1) + 1;
end
if nd2 ~= 0
    G(nd2,xr) = G(nd2,xr) - 1;
    G(xr,nd2) = G(xr,nd2) - 1;
end

if ni1 ~= 0
    G(xr,ni1) = G(xr,ni1) - val;
end
if ni2 ~= 0
    G(xr,ni2) = G(xr,ni2) + val;
end
end
```

1.8 Operatoinal Amplifier

OpAmp with infinite gain behave very similar to VCVS. The formula used is shown below:

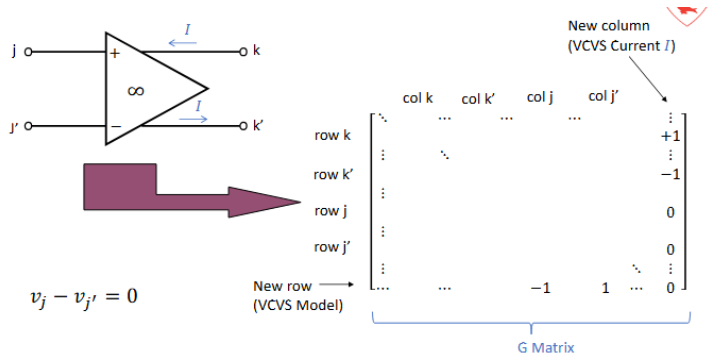

Figure 7: OpAmp Formula

```
function opa(inplus, inminus, out)
global G
global b
global C

d = size(G,1);      % current size
xr = d+1;           % new row index

b(xr) = 0;          % expand b
G(xr,xr) = 0;       % expand G
C(xr,xr) = 0;       % expand C

if out ~= 0
    G(out, xr) = G(out, xr) - 1;
end


if inplus ~= 0
    G(xr, inplus) = G(xr, inplus) - 1;
end

if inminus ~= 0
    G(xr, inminus) = G(xr, inminus) + 1;
end

end %end of function
```

1.9 Sample Circuit 1

By inputing the netlist for sample circuit one. The result of final G, C, b matrix is shown below.

$$\mathbf{G} = \begin{bmatrix} 2.0\times10^{-2} & -2.0\times10^{-2} & 0 & 0 & 1 & 0 & 0 & 0 \\ -2.0\times10^{-2} & 2.0\times10^{-2} & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 2.0\times10^{-2} & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3.190\times10^{-7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 6.372\times10^{-11} & -6.372\times10^{-11} & 0 & 0 & 0 & 0 \\ 0 & 0 & -6.372\times10^{-11} & 3.1906\times10^{-7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -3.170\times10^{-7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1.590\times10^{-6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -3.170\times10^{-7} \end{bmatrix}$$

$$\mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

The complete algorithm is show below

```
clear;
clc;
global G
global C
global b
G = zeros(4,4);
C = zeros(4,4);
b = zeros(4,1);
vol(1,0,1);
res(1,2,50);
res(4,0,50);
cap(2,0,0.319e-6);
cap(3,4,63.72e-12);
cap(4,0, 0.319e-6);
ind(2, 0, 0.317e-6);
ind(2, 3, 1.59e-6);
ind(4, 0, 0.317e-6);
```

2 Frequency Response

2.1 fsolve

Obtaining the G, C, b from the function in section 1, we then solve a system of linear equations at each frequency point using the formula below.

$$(G + j\omega_0 C)\, X(j\omega_0) = B$$

The code is displayed below:

```
function [fpoints, r] = fsolve(fL, fH, N, out)
global G C b
% log spacing
fpoints = logspace(log10(fL), log10(fH), N);
r = zeros(N,1);
for k = 1:N
    omega = 2*pi*fpoints(k);
    A = G + 1j*omega*C;
    X = A \ b;
    r(k) = X(out);
end
end
```

2.2 Sample Circuit 2

2.2.1 Result Using OpAmp

G matrix is shown below

```
Columns 1 through 10

 1.0410e-04  -1.0410e-04           0           0           0           0           0           0           0           0
-1.0410e-04   2.9412e-04  -1.4706e-04           0           0           0           0           0           0           0
         0  -1.4706e-04   1.4706e-04           0           0           0           0           0           0           0
         0           0           0   1.0748e-04  -1.0748e-04           0           0           0           0           0
         0           0           0  -1.0748e-04   2.3415e-04  -1.9191e-05           0           0           0  -1.0748e-04
         0           0           0           0  -1.9191e-05   1.2667e-04  -1.0748e-04           0           0           0
         0           0           0           0           0  -1.0748e-04   1.0748e-04           0           0           0
         0           0           0           0           0           0           0   5.0000e-05  -5.0000e-05           0
         0           0           0           0           0           0           0  -5.0000e-05   1.0000e-04  -5.0000e-05
         0           0           0           0  -1.0748e-04           0           0           0  -5.0000e-05   1.5748e-04
 1.0000e+00           0           0           0           0           0           0           0           0           0
         0           0  -1.0000e+00   1.0000e+00           0           0           0           0           0           0
         0           0           0           0   1.0000e+00           0           0           0           0           0
         0           0           0           0           0           0   1.0000e+00           0           0           0
         0           0           0           0           0           0           0           0   1.0000e+00           0

Columns 11 through 15

 1.0000e+00           0           0           0           0
         0           0           0           0           0
         0           0           0           0           0
         0  -1.0000e+00           0           0           0
         0           0           0           0           0
         0           0  -1.0000e+00           0           0
         0           0           0           0           0
         0           0           0  -1.0000e+00           0
         0           0           0           0           0
         0           0           0           0  -1.0000e+00
         0           0           0           0           0
         0           0           0           0           0
         0           0           0           0           0
         0           0           0           0           0
         0           0           0           0           0
```

Figure 8: G Matrix OpAmp

C matrix is shown below

```
Columns 1 through 10

         0         0         0         0         0         0         0         0         0
         0    9.4900e-08         0   -9.4900e-08         0         0         0         0         0
         0         0    2.0500e-08         0         0         0         0         0         0
         0   -9.4900e-08         0    9.4900e-08         0         0         0         0         0
         0         0         0         0    1.5000e-08   -1.5000e-08         0         0         0
         0         0         0         0   -1.5000e-08    1.5000e-08         0         0         0
         0         0         0         0         0         0    1.5000e-08   -1.5000e-08         0
         0         0         0         0         0         0   -1.5000e-08    1.5000e-08         0
         0         0         0         0         0         0         0         0         0
         0         0         0         0         0         0         0         0         0
         0         0         0         0         0         0         0         0         0
         0         0         0         0         0         0         0         0         0
         0         0         0         0         0         0         0         0         0
         0         0         0         0         0         0         0         0         0
         0         0         0         0         0         0         0         0         0

Columns 11 through 15

         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
         0         0         0         0         0
```

Figure 9: C Matrix OpAmp

b matrix is shown below

$$
\begin{matrix}
0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0
\end{matrix}
$$

Figure 10: b Matrix OpAmp

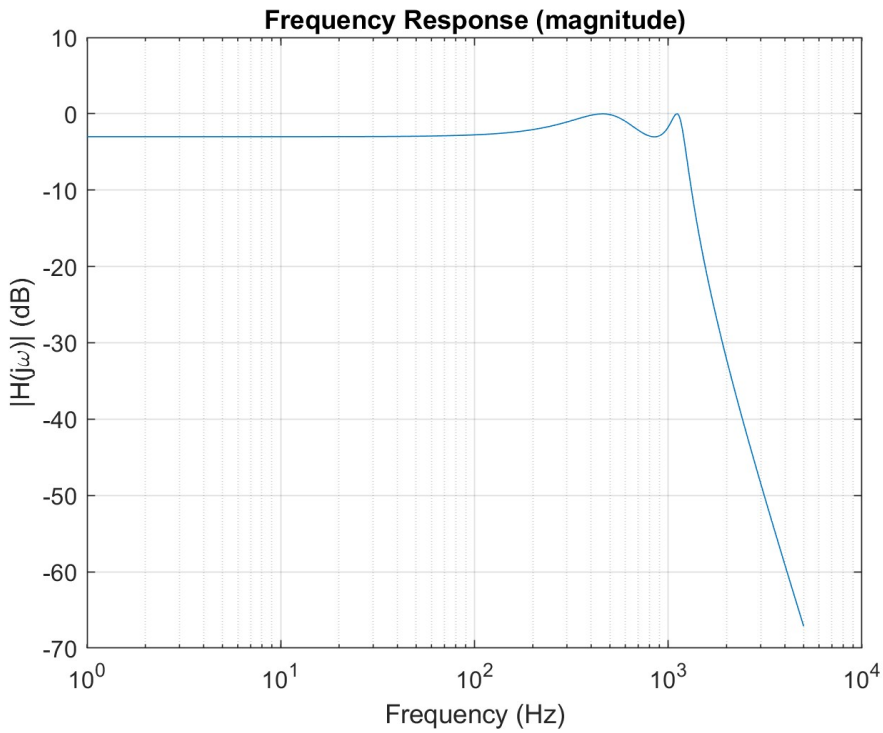

Figure 11: OpAmp Magnitude

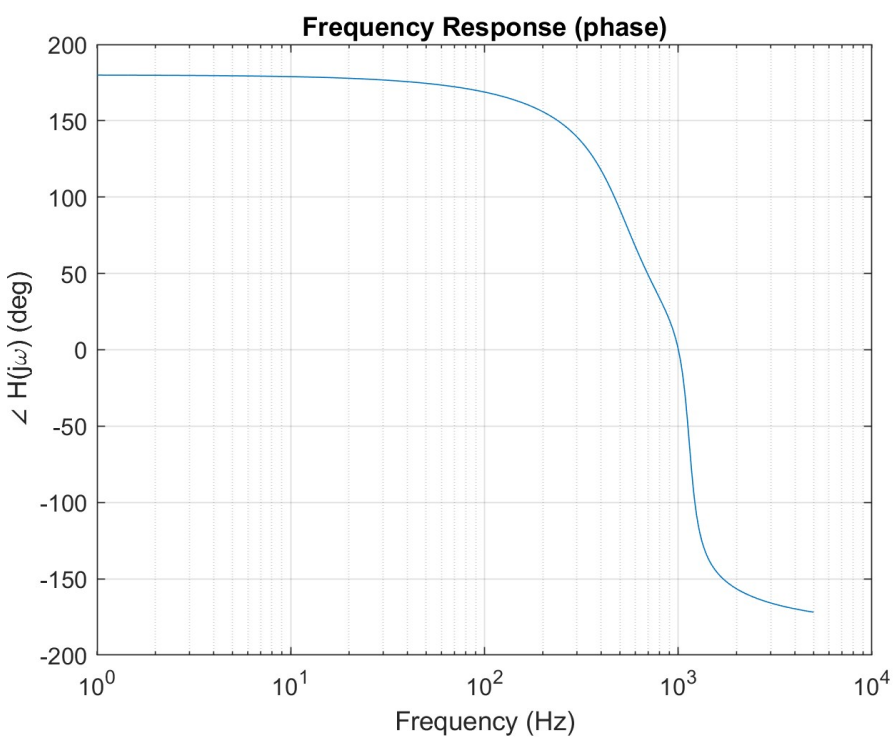

Figure 12: OpAmp Phase

2.2.2 Result Using VCVS

G matrix is shown below

```
Columns 1 through 11

 1.0410e-04  -1.0410e-04           0           0           0           0           0           0           0           0   1.0000e+00
-1.0410e-04   2.9412e-04  -1.4706e-04           0           0           0           0           0           0           0           0
          0  -1.4706e-04   1.4706e-04           0           0           0           0           0           0           0           0
          0           0           0   1.0748e-04  -1.0748e-04           0           0           0           0           0           0
          0           0           0  -1.0748e-04   2.3415e-04  -1.9191e-05           0           0           0  -1.0748e-04           0
          0           0           0           0  -1.9191e-05   1.2667e-04  -1.0748e-04           0           0           0           0
          0           0           0           0           0  -1.0748e-04   1.0748e-04           0           0           0           0
          0           0           0           0           0           0           0   5.0000e-05  -5.0000e-05           0           0
          0           0           0           0           0           0           0  -5.0000e-05   1.0000e-04  -5.0000e-05           0
          0           0           0           0  -1.0748e-04           0           0           0  -5.0000e-05   1.5748e-04           0
 1.0000e+00           0           0           0           0           0           0           0           0           0           0
          0           0  -5.0000e+01   4.9000e+01           0           0           0           0           0           0           0
          0           0           0           0   5.0000e+01  -1.0000e+00           0           0           0           0           0
          0           0           0           0           0           0   5.0000e+01  -1.0000e+00           0           0           0
          0           0           0           0           0           0           0           0   5.0000e+01  -1.0000e+00           0

Columns 12 through 15

          0           0           0           0
          0           0           0           0
          0           0           0           0
-1.0000e+00           0           0           0
          0           0           0           0
          0  -1.0000e+00           0           0
          0           0           0           0
          0           0  -1.0000e+00           0
          0           0           0           0
          0           0           0  -1.0000e+00
          0           0           0           0
          0           0           0           0
          0           0           0           0
          0           0           0           0
          0           0           0           0
```

Figure 13: G Matrix VCVS

C matrix is shown below

```
Columns 1 through 11

          0           0           0           0           0           0           0           0           0           0           0
          0   9.4900e-08           0  -9.4900e-08           0           0           0           0           0           0           0
          0           0   2.0500e-08           0           0           0           0           0           0           0           0
          0  -9.4900e-08           0   9.4900e-08           0           0           0           0           0           0           0
          0           0           0           0   1.5000e-08  -1.5000e-08           0           0           0           0           0
          0           0           0           0  -1.5000e-08   1.5000e-08           0           0           0           0           0
          0           0           0           0           0           0   1.5000e-08  -1.5000e-08           0           0           0
          0           0           0           0           0           0  -1.5000e-08   1.5000e-08           0           0           0
          0           0           0           0           0           0           0           0           0           0           0
          0           0           0           0           0           0           0           0           0           0           0
          0           0           0           0           0           0           0           0           0           0           0
          0           0           0           0           0           0           0           0           0           0           0
          0           0           0           0           0           0           0           0           0           0           0
          0           0           0           0           0           0           0           0           0           0           0
          0           0           0           0           0           0           0           0           0           0           0

Columns 12 through 15

          0           0           0           0
          0           0           0           0
          0           0           0           0
          0           0           0           0
          0           0           0           0
          0           0           0           0
          0           0           0           0
          0           0           0           0
          0           0           0           0
          0           0           0           0
          0           0           0           0
          0           0           0           0
          0           0           0           0
          0           0           0           0
          0           0           0           0
```

Figure 14: C Matrix VCVS

b matrix is shown below

$$
\begin{matrix}
0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0
\end{matrix}
$$

Figure 15: b Matrix VCVS

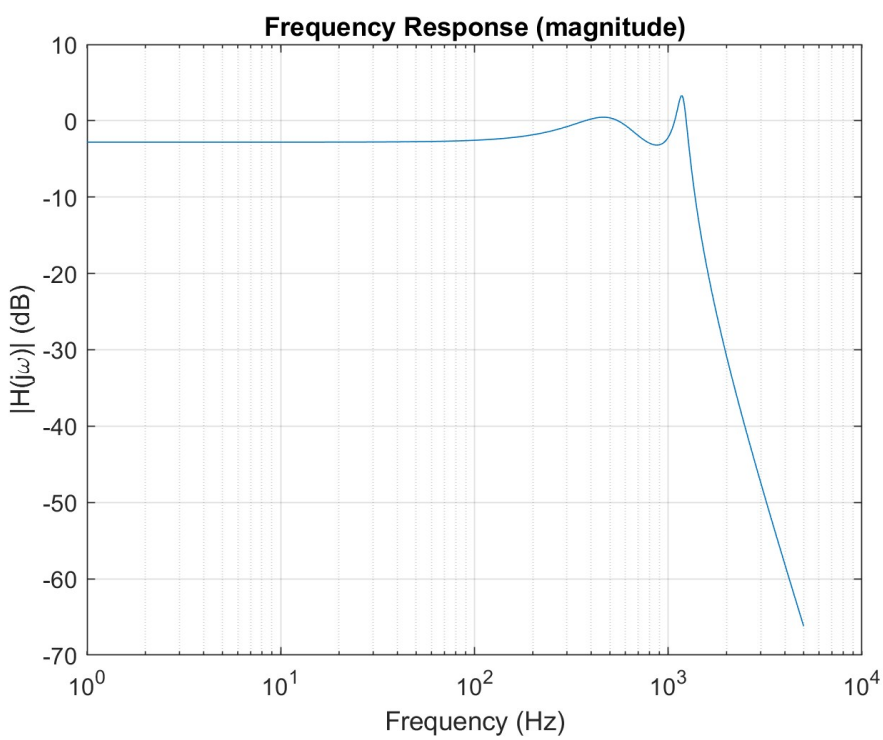

Figure 16: VCVS Magnitude

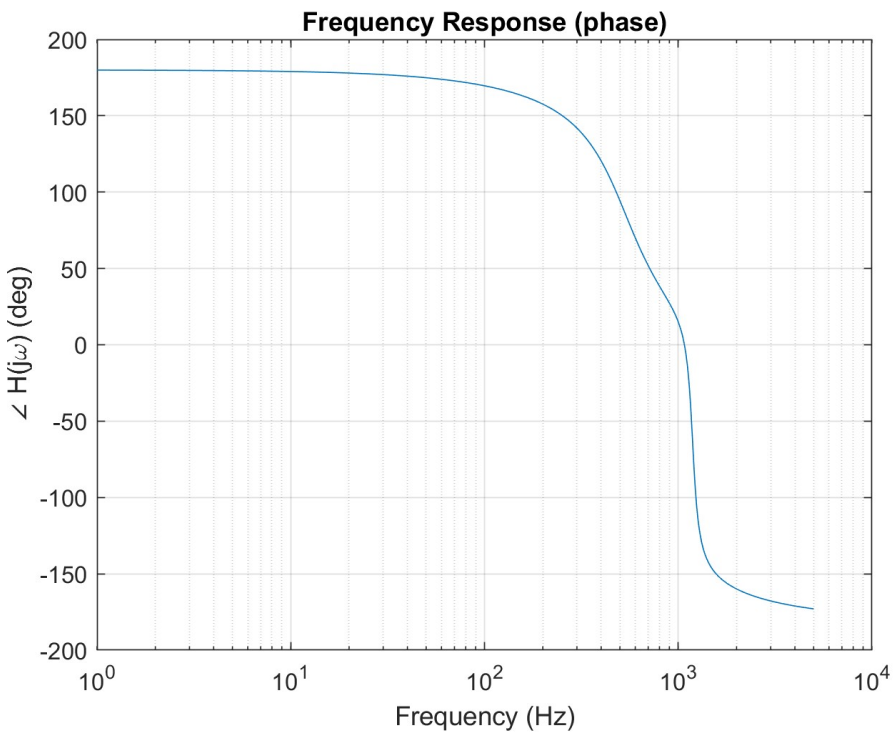

Figure 17: VCVS Phase

2.2.3 Code

The complete code is shown below:

```
clear global G C b;
clc;
global G
global C
global b
G = zeros(10,10);
C = zeros(10,10);
b = zeros(10,1);

% Source
vol(1,0,1);    % vi = cos(wt)

% Stage 1
res(1,2,9606);              % R1a
res(2,0,23280);             % R1b
res(2,3,6800);              % R2
cap(3,0,20.5e-9);           % C2
cap(2,4,94.9e-9);           % C1 (feedback)
%opa(3,4,4);                 % opamp1 ( + at node3, - at node2 )
vcvs(0,4,3,4,50);
```

```
% Stage 2
res(4,5,9304);              % Rg
res(5,6,52107);             % Rq (feedback)
cap(5,6,15e-9);             % C3 (feedback)
%opa(0,5,6);                 % opamp2 (+ grounded)
vcvs(0,6,0,5,50);
res(5,10,9304);             % R3 (from stage2 to final output)

% Stage 3
res(6,7,9304);              % R4
cap(7,8,15e-9);             % C4 (feedback)
%opa(0,7,8);                 % opamp3 (+ grounded)
vcvs(0,8,0,7,50);

%Stage 4
res(8,9,20000);             % r (input)
res(9,10,20000);            % r (feedback)

%opa(0,9,10);                % opamp4 (+ grounded)
vcvs(0,10,0,9,50);
format short e
disp(G)
disp(C)
disp(b)
% Output node
% Vout = node 10
fL = 1;
fH = 5000;
N = 1000;
out = 10;
[f, H] = fsolve(fL, fH, N, out);
mag_dB = 20*log10(abs(H));
phase_deg = unwrap(angle(H))*180/pi;
disp(20*log10(abs(H(1))));

figure;
semilogx(f, mag_dB);
grid on;
xlabel('Frequency (Hz)');
ylabel('|H(j\omega)| (dB)');
title('Frequency Response (magnitude)');

figure;
semilogx(f, phase_deg);
grid on;
xlabel('Frequency (Hz)');
ylabel('\angle H(j\omega) (deg)');
title('Frequency Response (phase)');
```